# Lec 03 Bellman Optimality Equation


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# Motivating examples

# Definition of optimal policy

The state value could be used to evaluate if a policy is good or not: if

$$v_{\pi_1}(s) \geq v_{\pi_2}(s) \quad \text{for all } s \in \mathcal{S}$$

then $\pi_1$ is better than $\pi_2$.

> 🏷️ **Definition**
> A policy $\pi^*$ is optimal if $v_{\pi^*}(s) \geq v_\pi(s)$ for all $s$ and for any other policy $\pi$.

# BOE: Introduction

## Bellman optimality equation (elementwise form)

$$
\begin{aligned}
v(s) &= \max_{\pi} \sum_{a} \pi(a|s) \left( \sum_{r} p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v(s') \right), \quad \forall s \in \mathcal{S} \\
&= \max_{\pi} \sum_{a} \pi(a|s) q(s,a) \quad s \in \mathcal{S}
\end{aligned}
$$

**Remarks:**

- $p(r|s,a), p(s'|s,a)$ are known.
- $v(s), v(s')$ are unknown and to be calculate.
- $\pi(s)$ here is unknown and to be solved.

## Bellman optimality equation (matrix-vector form):

$$
v = \max_{\pi} (r_\pi + \gamma P_\pi v)
$$

where the elements corresponding to $s$ or $s'$ are:

$$
[r_\pi]_s \triangleq \sum_{a} \pi(a|s) \sum_{r} p(r|s,a) r
$$

$$
[P_\pi] s, s' = p(s'|s) \triangleq \sum_{a} \pi(a|s) \sum s' p(s'|s,a)
$$

Here $\max_\pi$ is performed elementwise.

# Maximization on the RHS side of BOE(Elementwise form)

Fix $v'(s)$ first and solve $\pi$:

$$
\begin{aligned}
v(s) &= \max_{\pi} \sum_{a} \pi(a|s) \left( \sum_{r} p(r|s,a) r + \gamma \sum_{s'} p(s'|s,a) v(s') \right), \quad \forall s \in \mathcal{S} \\
&= \max_{\pi} \sum_{a} \pi(a|s) q(s,a) \quad s \in \mathcal{S}
\end{aligned}
$$

Consider the following example:

**Example (How to solve $\max_\pi \sum_a \pi(a|s)q(s,a)$)**

Suppose $q_1, q_2, q_3 \in \mathbb{R}$ are given. Find $c_1^*, c_2^*, c_3^*$ solving

$$\max_{c_1,c_2,c_3} \; c_1q_1 + c_2q_2 + c_3q_3.$$

where $c_1 + c_2 + c_3 = 1$ and $c_1, c_2, c_3 \geq 0$.

Without loss of generality, suppose $q_3 \geq q_1, q_2$. Then, the optimal solution is $c_3^* = 1$ and $c_1^* = c_2^* = 0$. That is because for any $c_1, c_2, c_3$

$$q_3 = (c_1 + c_2 + c_3)q_3 = c_1q_3 + c_2q_3 + c_3q_3 \geq c_1q_1 + c_2q_2 + c_3q_3.$$

Inspired by the example above, considering that $\sum_a \pi(a|s) = 1$, we have

$$\max_\pi \sum_a \pi(a|s)q(s,a) = \max_{a\in\mathcal{A}(s)} q(s,a)$$

where the optimality is achieved when

$$\pi(a|s) = \begin{cases} 1 & a = a^* \\ 0 & a \neq a^* \end{cases}$$

where $a^* = \arg\max_a q(s,a)$

# Matrix-vector form

Rewrite BOE as:

$$v = f(v) = \max_\pi (r_\pi + \gamma P_\pi v)$$

## Contraction mapping theorem

- **Fixed point**: $x \in X$ is a fixed point of $f : X \to X$ if

$$f(x) = x$$

- **Contraction mapping (or contractive function)**: $f$ is a contraction mapping if

$$\|f(x_1) - f(x_2)\| \leq \gamma \|x_1 - x_2\|$$

where $\gamma \in (0, 1)$.

  - $\gamma$ must be strictly less than $1$ so that many limits such as $\gamma^k \to 0$ as $k \to \infty$ hold.
  - Here$\|\cdot\|$ can be any vector norm.

**Theorem (Contraction Mapping Theorem)**

*For any equation that has the form of $x = f(x)$, if $f$ is a contraction mapping, then*

- *Existence: there exists a fixed point $x^*$ satisfying $f(x^*) = x^*$.*
- *Uniqueness: The fixed point $x^*$ is unique.*
- *Algorithm: Consider a sequence $\{x_k\}$ where $x_{k+1} = f(x_k)$, then $x_k \to x^*$ as $k \to \infty$. Moreover, the convergence rate is exponentially fast.*

# Solution

- For BOE: $v = f(v) = \max_\pi(r_\pi + \gamma P_\pi v)$, $f(v)$ is a **contraction mapping** satisfying

$$\|f(x_1) - f(x_2)\| \leq \gamma \|x_1 - x_2\|$$

where $\gamma$ is the discount rate.

- Therefore, BOE always exists a unique fixed point $v^*$, and the solution can be sovled iteractively by

$$v_{k+1} = f(v_k) = \max_\pi(r_\pi + \gamma P_\pi v_k)$$

This sequence $\{v_k\}$ converges to $v^*$ exponentially fast given any initial guess $v_0$. The convergence rate is determined by $\gamma$.

- Suppose

$$\pi^* = \arg\max_{\pi}(r_\pi + \gamma P_\pi v^*)$$

Then

$$v^* = r_{\pi^*} + \gamma P_{\pi^*} v^*$$

Therefore, $\pi^*$ is a policy and $v^* = v_{\pi^*}$ is the corresponding state value.

## Policy optimality

**Theorem (Policy Optimality)**

*Suppose that $v^*$ is the unique solution to $v = \max_\pi(r_\pi + \gamma P_\pi v)$, and $v_\pi$ is the state value function satisfying $v_\pi = r_\pi + \gamma P_\pi v_\pi$ for any given policy $\pi$, then*

$$v^* \geq v_\pi, \quad \forall \pi$$

Therefore $\pi^*$ is the optimal policy. It is a deterministic greedy policy as below:

$$\pi^*(a|s) = \begin{cases} 1 & a = a^*(s) \\ 0 & a \neq a^*(s) \end{cases}$$

where $a^*(s) = \arg\max_a q^*(a, s), q^*(s, a) := \sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v^*(s')$

# Analyzing optimal policies

- The optimal policies are invariant to the linear transformation of the reward signals.
- The optimal state value is unique, but the optimal policy may not be unique.